Concurrency sensitivity (Tier 2, item 35)

Every measurement in this project before now was single-query-at-a-time. Production is not. This measures runtime under N simultaneous clients of the same query, N in {1,2,4,8}, on the Apple M1 (8 cores).

`e_concurrency` is the log-log slope of runtime against client count. 0 means perfect scaling (the machine absorbs the extra load for free); **1 means fully serialised** (8 clients take 8x as long each, so throughput is flat).

Aggregate

clients	total throughput (queries/s)	median query slowdown vs 1 client
1	74.04	1.00x
2	87.06	2.02x
4	126.99	2.82x
8	153.01	4.69x

Median concurrency elasticity across queries: **+0.752** (range +0.286 to +0.984).

Partial scaling. Throughput rises with clients but sub-linearly, so contention is real but not total. This is the awkward middle for a predictor: the slowdown depends on the query, so a single global contention factor will not do.

Per query

query	e_concurrency	t at 1 client (ms)	slowdown at 8 clients	distinct plans
q11	+0.984	130	8.57x	1
q15	+0.931	743	6.53x	1
q4	+0.895	185	4.91x	1
q1	+0.877	2247	6.25x	1
q12	+0.863	582	6.09x	1
q6	+0.863	404	5.89x	1
q3	+0.856	370	5.12x	1
q9	+0.835	1719	5.78x	1
q13	+0.793	1346	5.90x	1
q5	+0.768	478	5.05x	1
q10	+0.763	364	5.20x	1
q14	+0.742	374	6.09x	1
q2	+0.695	599	4.27x	1
q16	+0.635	367	3.84x	1
q21	+0.628	1472	3.68x	1
q18	+0.581	3048	3.18x	1
q22	+0.546	57	3.57x	1
q19	+0.482	64	2.94x	1
q17	+0.435	1056	2.52x	1
q8	+0.327	335	2.00x	1
q7	+0.293	2611	1.83x	1
q20	+0.286	1406	1.88x	1

Sanity check: does the plan change with concurrency?

0 of 22 queries show more than one plan signature across concurrency levels. Concurrency is not a planner input, so the expected answer is zero.

Limitations

- **One machine, and self-contention only** — N clients all run the *same* query. A real mixed workload contends differently, and that is not measured here.

- 8 clients on 8 cores with `max_parallel_workers_per_gather=2` means the requested parallelism alone exceeds the core count, so some of the slowdown is scheduler queueing rather than resource contention.
- 3 measured repetitions per cell, warmup discarded.